

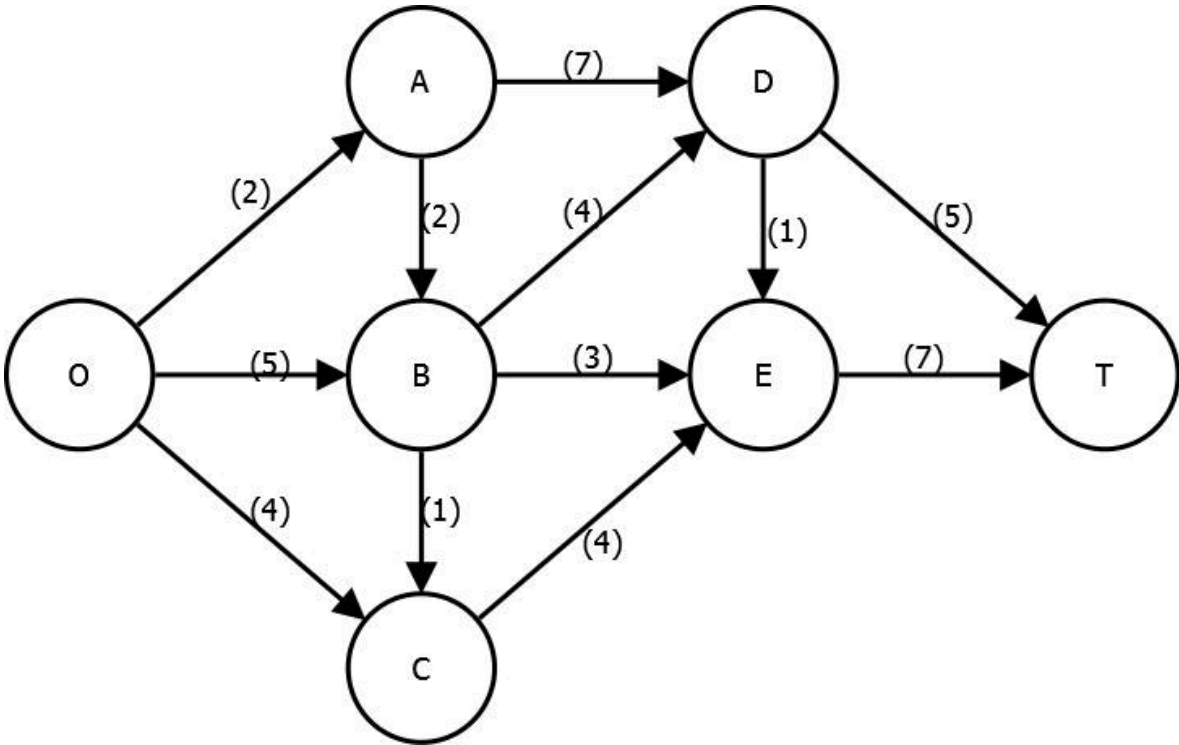
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1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#) [↗](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)
- 1 / 1 point

Consider a directed network depicted below.



Let the arc weights be lengths. For the following statements, select all that are correct.

- ☒ The path $(0, B, A, D, T)$ does not exist.
- ☒ Correct
- ☐ The total length of the path $(0, A, B, D, T)$ is 10.
- ☒ The total length of the path $(0, A, B, D, T)$ is 13.
- ☒ Correct
- ☒ There is no cycle in this graph.
- ☒ Correct
- ☐ None of the above.

2. Continue from the previous question. What is the total distance of a shortest path?
- 1 / 1 point
- ☐ 12.
- ☒ 13.
- ☐ 14.
- ☐ 15.
- ☐ None of the above.
- ☒ Correct

3. Continue from Question 1. Let the arc weights be flow capacities. What is the flow value of a maximum flow solution?
- 1 / 1 point
- ☐ 10.
- ☒ 11.
- ☐ 12.
- ☐ 13.
- ☐ None of the above.
- ☒ Correct

4. Continue from the previous question. Which of the following is not possible in a maximum flow solution?
- 1 / 1 point
- ☐ $x_{DT} = 4$.
- ☐ $x_{ET} = 7$.
- ☒ $x_{BC} = 1$.
- ☐ $x_{DE} = 1$.
- ☐ None of the above.
- ☒ Correct

5. Consider the following matrices
- 1 / 1 point
- $$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$
- ☐ A is totally unimodular but B is not.
- ☐ B is totally unimodular but A is not.
- ☒ Both A and B are totally unimodular.
- ☐ Both A and B are not totally unimodular.
- ☐ None of the above.
- ☒ Correct